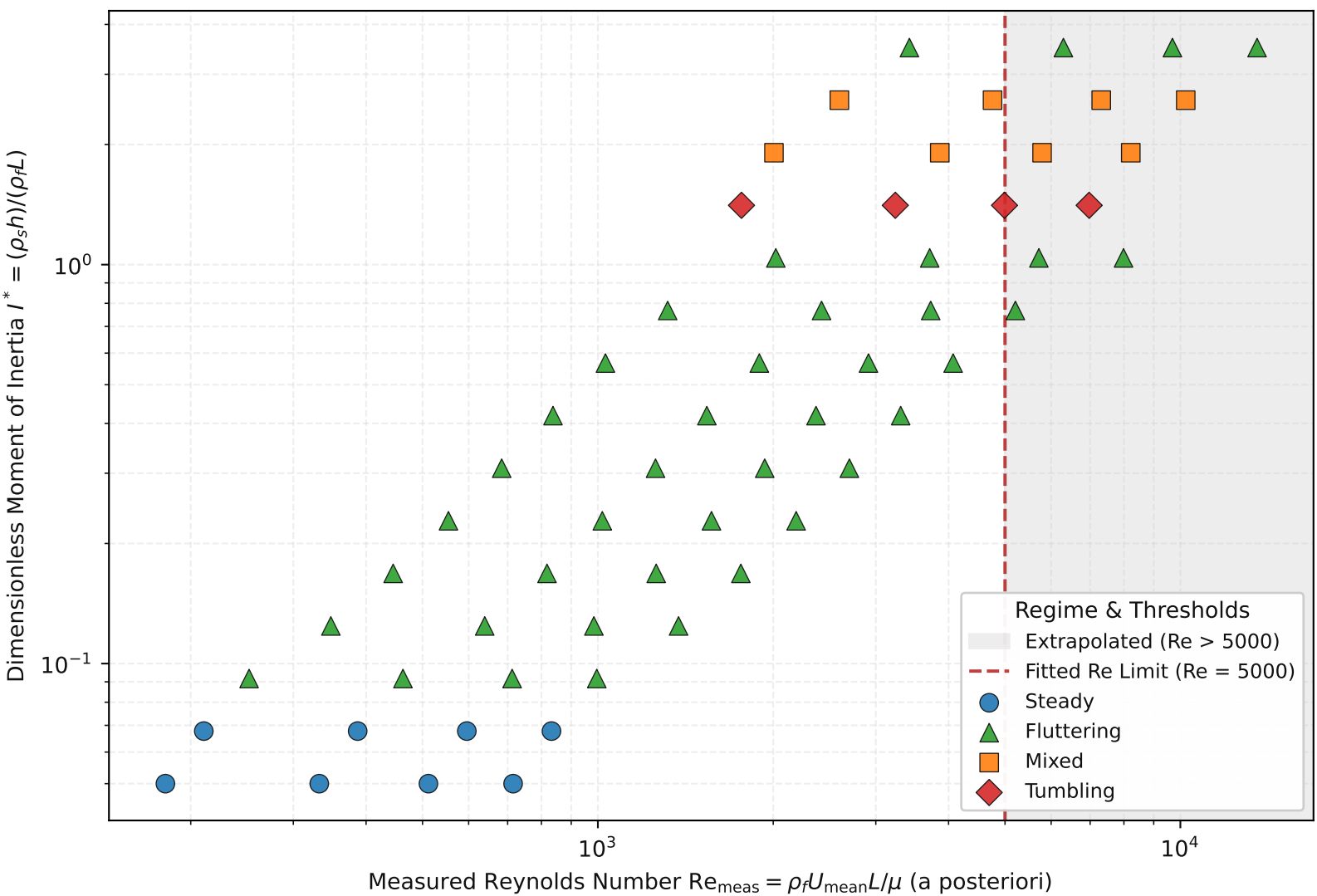
Phase Diagram: Falling Regime Map ($I^*$ vs Re) (2D quasi-steady model simulation)
Dimensionless Moment of Inertia $I^* = (\rho_s h)/(\rho_f L)$
$10^0$
$10^{-1}$
$10^3$
$10^4$
Measured Reynolds Number $\mathrm{Re}_{\mathrm{meas}} = \rho_f U_{\mathrm{mean}} L/\mu$ (a posteriori)
Regime & Thresholds
Extrapolated (Re > 5000)
Fitted Re Limit (Re = 5000)
Steady
Fluttering
Mixed
Tumbling